



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

Chip Design Verification - ENCS5337

Homework 4

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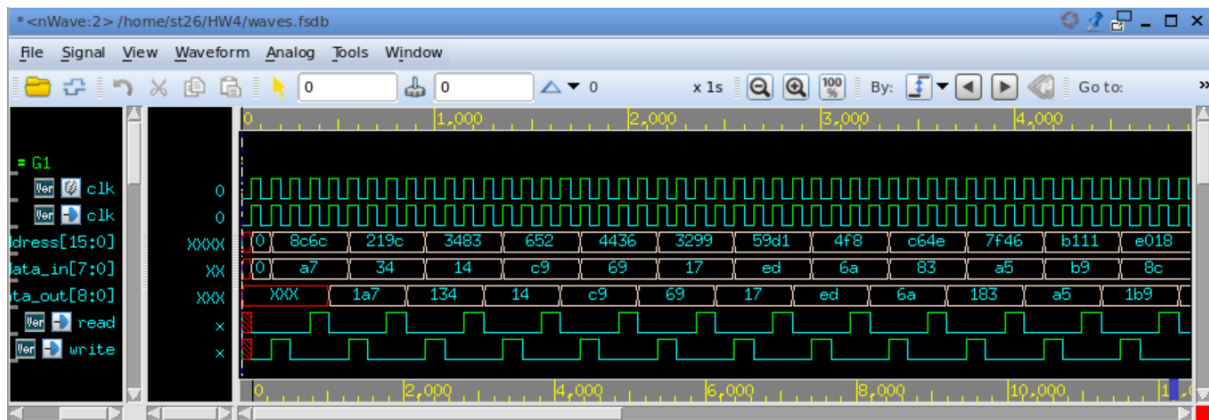
Section: 1

Instructor: Dr. Elias Khalil

May 3rd 2026

Deliverables

1. Linux Path to the source code folder on the Synopsys server
/home/st26/HW4
2. Waveform screenshot



3. Log files

/home/st26/HW4/compile.log

```
Open compile.log Save
~/HW4
Command: vcs -full64 -sverilog -debug_access+all -kdb -l compile.log -f flist.f
Chronologic VCS (TM)
Version T-2022.06-SP2_Full64 -- Sun May 3 02:54:19 2026

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Parsing design file 'mem_pkg.sv'
Parsing design file 'mem_if.sv'
Parsing design file 'my_mem.sv'
Parsing design file 'program_test.sv'
Parsing design file 'top.sv'
Top Level Modules:
    top
No TimeScale specified
Notice: Ports coerced to inout, use -notice for details
Starting vcs inline pass...

4 modules and 0 UDP read.
recompiling module mem_if
recompiling module test
Both modules done.
    However, due to incremental compilation, only 2 modules need to be compiled.
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so
```

```

        However, due to incremental compilation, only 2 modules need to be compiled.
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so
if [ -x ../simv ]; then chmod a-x ../simv; fi
g++ -o ../simv -rdynamic -Wl,-rpath='$ORIGIN'/simv.daidir -Wl,-rpath=../simv.daidir \
-Wl,-rpath=/usr/synopsys/vcs/T-2022.06-SP2/linux64/lib -L/usr/synopsys/vcs/T-2022.06-SP2/linux64/
lib \
-Wl,-rpath-link=../usr/lib64/libnuma.so.1 objs/amcQw_d.o _34916_archive_1.so \
_prev_archive_1.so SIM_l.o rmapats_mop.o rmapats.o rmar.o rmar_nd.o rmar_llvm_0_1.o \
rmar_llvm_0_0.o -lvirsim -lerrorinf -lsnpsmalloc -lvfs -lvcsnew -lsimprofile \
-lucinatative /usr/synopsys/vcs/T-2022.06-SP2/linux64/lib/vcs_tls.o -Wl,-whole-archive \
-lvcsucli -Wl,-no-whole-archive _vcs_pli_stub.o /usr/synopsys/vcs/T-2022.06-SP2/
linux64/lib/vcs_save_restore_new.o \
/usr/synopsys/verdi/T-2022.06-SP2/share/PLI/VCS/LINUX64/pli.a -ldl -lc -lm -lpthread \
-ldl
../simv up to date
CPU time: .679 seconds to compile + .434 seconds to elab + .330 seconds to link
Verdi KDB elaboration done and the database successfully generated: 0 error(s), 0 warning(s)

```

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/home/st26/HW4/run.log

Open  run.log ~/HW4 Save  -  x

```

Command: /home/st26/HW4/./simv -l run.log
Chronologic VCS simulator copyright 1991-2022
Contains Synopsys proprietary information.
Compiler version T-2022.06-SP2_Full64; Runtime version T-2022.06-SP2_Full64; May 3 02:54 2026
*Verdi* Loading libsscore_vcs202206.so
FSDB Dumper for VCS, Release Verdi_T-2022.06-SP2, Linux x86_64/64bit, 11/22/2022
(C) 1996 - 2022 by Synopsys, Inc.
*Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the programs that are using this file.
*Verdi* : Create FSDB file 'waves.fsdb'
*Verdi* : Begin traversing the scope (top), layer (0).
*Verdi* : End of traversing.
*Verdi* : Begin traversing the SVA assertions, layer (0).
*Verdi* : End of traversing the SVA assertions.
[150] GEN: addr=8c6c data_in=a7
[250] GEN: addr=219c data_in=34
[350] GEN: addr=3483 data_in=14
[450] GEN: addr=652 data_in=c9
[550] GEN: addr=4436 data_in=69
[550] DRV: addr=8c6c data_in=a7 | expected=110100111
[550] data_out = 110100111 (hex: 1a7)
[550] PASS: addr=8c6c | data=110100111
[650] GEN: addr=3299 data_in=17
[750] GEN: addr=59d1 data_in=ed
[850] GEN: addr=4f8 data_in=6a
[950] GEN: addr=c64e data_in=83
[950] DRV: addr=219c data_in=34 | expected=100110100
[950] data_out = 100110100 (hex: 134)
[950] PASS: addr=219c | data=100110100
[1050] GEN: addr=7f46 data_in=a5
[1150] GEN: addr=b111 data_in=b9
[1250] GEN: addr=e018 data_in=8c
[1350] GEN: addr=f487 data_in=47
[1350] DRV: addr=3483 data_in=14 | expected=10100
[1350] data_out = 10100 (hex: 14)
[1350] PASS: addr=3483 | data=10100
[1450] GEN: addr=4ebc data_in=83
[1550] GEN: addr=28d3 data_in=c2
[1650] GEN: addr=868e data_in=f9
[1750] GEN: addr=3a1e data_in=b0
[1750] DRV: addr=652 data_in=c9 | expected=1101001

```

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Open

run.log
~/HW4

Save

```
[2150] DRV: addr=4436 data_in=69 | expected=1101001
[2150] data_out = 1101001 (hex: 69)
[2150] PASS: addr=4436 | data=1101001
[2550] DRV: addr=3299 data_in=17 | expected=10111
[2550] data_out = 10111 (hex: 17)
[2550] PASS: addr=3299 | data=10111
[2950] DRV: addr=59d1 data_in=ed | expected=11101101
[2950] data_out = 11101101 (hex: ed)
[2950] PASS: addr=59d1 | data=11101101
[3350] DRV: addr=4f8 data_in=6a | expected=1101010
[3350] data_out = 1101010 (hex: 6a)
[3350] PASS: addr=4f8 | data=1101010
[3750] DRV: addr=c64e data_in=83 | expected=110000011
[3750] data_out = 110000011 (hex: 183)
[3750] PASS: addr=c64e | data=110000011
[4150] DRV: addr=7f46 data_in=a5 | expected=10100101
[4150] data_out = 10100101 (hex: a5)
[4150] PASS: addr=7f46 | data=10100101
[4550] DRV: addr=b111 data_in=b9 | expected=110111001
[4550] data_out = 110111001 (hex: 1b9)
[4550] PASS: addr=b111 | data=110111001
[4950] DRV: addr=e018 data_in=8c | expected=110001100
[4950] data_out = 110001100 (hex: 18c)
[4950] PASS: addr=e018 | data=110001100
[5350] DRV: addr=f487 data_in=47 | expected=1000111
[5350] data_out = 1000111 (hex: 47)
[5350] PASS: addr=f487 | data=1000111
[5750] DRV: addr=4ebc data_in=83 | expected=110000011
[5750] data_out = 110000011 (hex: 183)
[5750] PASS: addr=4ebc | data=110000011
[6150] DRV: addr=28d3 data_in=c2 | expected=111000010
[6150] data_out = 111000010 (hex: 1c2)
[6150] PASS: addr=28d3 | data=111000010
[6550] DRV: addr=868e data_in=f9 | expected=11111001
[6550] data_out = 11111001 (hex: f9)
[6550] PASS: addr=868e | data=11111001
[6950] DRV: addr=3a1e data_in=b0 | expected=110110000
[6950] data_out = 110110000 (hex: 1b0)
[6950] PASS: addr=3a1e | data=110110000
[7350] DRV: addr=2a5a data_in=d2 | expected=11010010

[7350] data_out = 11010010 (hex: d2)
[7350] PASS: addr=2a5a | data=11010010
[7750] DRV: addr=4aa7 data_in=ae | expected=110101110
[7750] data_out = 110101110 (hex: 1ae)
[7750] PASS: addr=4aa7 | data=110101110
[8150] DRV: addr=58fa data_in=d9 | expected=111011001
[8150] data_out = 111011001 (hex: 1d9)
[8150] PASS: addr=58fa | data=111011001
[12150] Total errors = 0
[12150] Simulation complete.
$finish called from file "program_test.sv", line 35.
$finish at simulation time 12150
V C S S i m u l a t i o n R e p o r t
Time: 12150
CPU Time: 0.370 seconds; Data structure size: 0.0Mb
Sun May 3 02:54:33 2026
```

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